

Cyber Security Lab Exercise - 2

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Roll No: 16

TASK 1: WHOIS/RDAP Analysis

University: du.ac.in

Field	Observed Value	Analyst Interpretation
Registrar	ERNET India	Government-backed academic network → typical for Indian universities
Registration Date	28 Feb 2004	Long-standing domain → stable and established institution
Expiry Date	28 Feb 2028	Long renewal period → good administrative planning
Last Updated	May 2024	Regular maintenance → active domain management
Name Servers	himalaya.du.ac.in, himalaya1.du.ac.in	Self-hosted DNS → internal infrastructure control
Registrant/Admin Contact	Delhi University Computer Centre (redacted)	Managed internally but sensitive details hidden
Privacy Protection	Enabled (data redacted)	Limits exposure of individuals
Domain Status	clientTransferProhibited, clientUpdateProhibited	Strong protection against unauthorized changes

Search the WHOIS Database

WHOIS search results

Domain Information ^

Name	du.ac.in
Registry Domain ID	D13836-IN
Registered On	2004-02-28T05:00:00.039Z
Expires On	2028-02-28T05:00:00.000Z
Updated On	2024-05-06T12:14:57.903Z
Domain Status	client update prohibited client transfer prohibited
Name Servers	himalaya1.du.ac.in himalaya.du.ac.in

Registrar Information ^

Name	ERNET India
IANA ID	800068
Abuse Contact Email	tejalt@eis.ernet.in
Abuse Contact Phone	tel:+91.1123358248

Registrant Contact ^

Name	-
Organization	Delhi University Computer Centre
Phone	-
Fax	-
Email	-
Mailing Address	-

NTPC - A Public sector organization

Field	Observed Value	Analyst Interpretation
Registrar	Netlynx Technologies Pvt. Ltd	Indian registrar → organisation uses domestic domain management services
Registration Date	28 Feb 2003	Very old domain → long-established digital presence
Expiry Date	28 Feb 2031	Moderate renewal period → requires timely renewal attention
Last Updated	March 2026	Indicates active maintenance and updates
Name Servers	ns-132.awsdns-16.com ns-1808.awsdns-34.co.uk ns-1030.awsdns-00.org ns-691.awsdns-22.net	Use of cloud-based DNS → scalable and reliable infrastructure
Registrant/Admin Contact	REDACTED FOR PRIVACY (NTPC LTD.)	Sensitive details hidden
Privacy Protection	Enabled	Reduces exposure of personnel data
Domain Status	clientTransferProhibited	Prevents unauthorized domain transfer

Search the WHOIS Database

https://ntpc.co.in/

WHOIS search results

Domain Information

Name	ntpc.co.in
Registry Domain ID	D7739-IN
Registered On	2003-02-28T05:00:00.720Z
Expires On	2031-02-28T05:00:00.000Z
Updated On	2026-03-03T13:29:59.514Z
Domain Status	client transfer prohibited
Name Servers	ns-132.awsdns-16.com ns-1808.awsdns-34.co.uk ns-1030.awsdns-00.org ns-691.awsdns-22.net

Registrar Information

Name	Netlynx Technologies Pvt. Ltd.
IANA ID	1004
Abuse Contact Email	susheel@netlynx.com
Abuse Contact Phone	tel:+91.2240811144

Registrant Contact

Name	-
Organization	NTPC LTD.
Phone	-
Fax	-
Email	-
Mailing Address	Delhi

Flipkart - a commercial organisation with a publicly visible web presence.

Field	Observed Value	Analyst Interpretation
Registrar	GoDaddy.com, LLC	Large global registrar → enterprise-level domain management
Registration Date	3 June 2007	Long-standing domain → mature digital presence
Expiry Date	3 June 2026	Regular renewal cycle → requires periodic monitoring
Last Updated	June 2025	Actively maintained domain
Name Servers	sdns14.ultradns.biz/com/net/org	Third-party DNS (UltraDNS) → highly scalable infrastructure
Registrant/Admin Contact	Domains By Proxy, LLC (Registration Private)	Identity hidden behind proxy service
Privacy Protection	Enabled (full masking)	Strong protection of ownership details
Domain Status	clientTransferProhibited, clientUpdateProhibited, etc.	Prevents unauthorized domain changes

WHOIS search results

Domain Information

Name	FLIPKART.COM
Registry Domain ID	1009529768_DOMAIN_COM-VRSN
Registered On	2007-06-03T19:32:20Z
Expires On	2026-06-03T19:32:20Z
Updated On	2025-06-04T11:59:40Z
Domain Status	client delete prohibited client renew prohibited client transfer prohibited client update prohibited
Name Servers	SDNS14.ULTRADNS.BIZ SDNS14.ULTRADNS.COM SDNS14.ULTRADNS.NET SDNS14.ULTRADNS.ORG

Registrant Contact

Name	Registration Private
Organization	Domains By Proxy, LLC
Phone	tel:+1.4806242599
Fax	—
Email	https://www.godaddy.com/whois/results.aspx?domain=FLIPKART.COM&action=contactDomainOwner
Mailing Address	DomainsByProxy.com 100 S. Mill Ave, Suite 1600, Tempe, Arizona, 85281

Technical Contact

Name	Registration Private
Organization	Domains By Proxy, LLC
Phone	tel:+1.4806242599
Fax	—
Email	https://www.godaddy.com/whois/results.aspx?domain=FLIPKART.COM&action=contactDomainOwner
Mailing Address	DomainsByProxy.com 100 S. Mill Ave, Suite 1600, Tempe, Arizona, 85281

Registrar Information

Name	GoDaddy.com, LLC
IANA ID	146
Abuse Contact Email	abuse@godaddy.com
Abuse Contact Phone	tel:+1.480-624-2505

1. What does the domain registration pattern suggest about the organisation's administrative maturity?

du.ac.in → usually government-controlled academic domain, moderate maturity

ntpc.co.in → public sector, likely structured and long-term registration

flipkart.com → highly mature, professionally managed, long expiry & renewals

Commercial organisations like flipkart.com show high administrative maturity with long-term registration and professional domain management, while government and university domains show stable but comparatively less optimized registration practices.

2. Does the presence or absence of privacy protection change the target's exposure? Explain.

Flipkart → likely uses privacy protection / corporate masking

du.ac.in / ntpc → often more transparent

Absence of privacy protection increases exposure by revealing administrative or technical contacts, making reconnaissance easier. Domains with privacy protection reduce direct targeting risk.

3. What does an expiry date in the near future tell an attacker about a domain?

A domain nearing expiry indicates potential vulnerability, as attackers may attempt domain hijacking or re-registration if renewal is missed.

4. If the registrant contact shows a named individual's personal email, what type of attack does this enable?

It enables phishing and social engineering attacks, where attackers can impersonate trusted entities to gain access or credentials.

TASK 2: (a) Shodan-based Exposure Assessment

Shodan Maps Images Monitor Developer More...

SHODAN Explore Pricing ↗ apache 🔍 Login

TOTAL RESULTS
16,084,847

TOP COUNTRIES



United States	4,241,384
Japan	1,505,058
Germany	1,503,534
China	765,053
France	660,205

[More...](#)

View Report Browse Images View on Map Advanced Search

Product Spotlight: A native desktop app to access Shodan using your API key. Fast and optimized for power users. Check out [Shodan Terminal](#)

400 Bad Request 🔗
103.204.135.53
53.135.204.103 jnpl.jeebr.net
Juwariyah Networks Private Limited
India, Mumbai
📌
2026-04-01T17:11:54.59941

HTTP/1.1 400 Bad Request
Date: Wed, 01 Apr 2026 17:07:42 GMT
Server: Apache/2.4.41 (Ubuntu)
Content-Length: 444
Connection: close
Content-Type: text/html; charset=iso-8859-1

302 Found 🔗
216.228.192.14
Zohy Fiber
United States, La Grande
📌
2026-04-01T17:10:25.65020

HTTP/1.1 302 Found
Date: Wed, 01 Apr 2026 17:06:12 GMT
Server: Apache/2.4.29 (Ubuntu)
Location: https://216.228.192.14/
Content-Length: 287
Content-Type: text/html; charset=iso-8859-1

TOP PORTS

80	5,917,300
443	4,686,964
8080	318,752
5006	134,485
8081	130,087

[More...](#)

TOP ORGANIZATIONS

Amazon Technologies Inc.	616,883
Amazon.com, Inc.	472,401
DigitalOcean, LLC	437,091
Hetzner Online GmbH	376,746
Aliyun Computing Co., LTD	305,237

[More...](#)

TOP PRODUCTS

Apache httpd	11,157,265
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WELCOME TO SCHOOLAUTOMATE 🔗
124.106.169.176
IPG
Philippines, Baguio
📌
2026-04-01T17:09:58.190083

HTTP/1.1 200 OK
ETag: W/"3721-1681182703921"
Last-Modified: Tue, 11 Apr 2023 03:11:43 GMT
Content-Type: text/html
Content-Length: 3721
Date: Wed, 01 Apr 2026 17:05:45 GMT
Server: Apache-Coyote/1.1

PaperCut Login 🔗
3.239.60.0
ec2-3-239-60-0.compute-1.amazonaws.com
ws.com
Amazon Data Services NoVa
United States, Ashburn
📌
2026-04-01T17:09:29.908707

HTTP/1.1 200 OK
Date: Wed, 01 Apr 2026 17:05:17 GMT
Server: Apache-AdvancedExtranetServer/2.0.47 (Mandrake Linux/6.12.92mdk) mod_perl/1.99.89 Perl/v5.8.1 mod_s
Last-Modified: Thu, 16 Jun 2016 14:46:34 GMT
Content-Type: text/html
X-Frame-Options: SAMEORIGIN
...

Apache HTTP Server Test Page powered by CentOS 🔗
80.202.22.12
SK Broadband Co Ltd
Korea, Republic of, Cheongju-si
📌
2026-04-01T17:09:21.808724

HTTP/1.1 403 Forbidden
Date: Wed, 01 Apr 2026 17:05:09 GMT
Server: Apache/2.4.6 (CentOS)
Last-Modified: Thu, 16 Oct 2014 13:20:58 GMT
ETag: "1321-5058a1e728288"
Accept-Ranges: bytes
Content-Length: 4897
Connection: close

Domain: flipkart.com

SHODAN Explore Pricing ↗ flipkart.com 🔍 Login

TOTAL RESULTS
10

TOP COUNTRIES



India	9
United States	1

TOP PORTS

443	6
80	3
53	1

View Report View on Map Advanced Search

Product Spotlight: Keep track of what you have connected to the Internet. Check out [Shodan Monitor](#)

Ekart Logistics 🔗
163.53.76.136
Flipkart Internet Pvt Ltd
India, Mumbai
📌
2026-04-01T16:43:42.36567

HTTP/1.1 200 OK
Set-Cookie: empId=; Path=/; Secure; HTTPOnly
cache-control: max-age=2592000
x-xss-protection: 1; mode=block
x-content-type-options: nosniff
content-security-policy: style-src 'self' 'unsafe-inline' https://flipkartads.azureedge.net https://fonts.googleapis.com/ ht

Flipkart - Demand Platform 🔗
103.243.32.24
Flipkart Internet Pvt Ltd
India, Chennai
📌
2026-04-01T16:28:09.670721

HTTP/1.1 200 OK
Server: nginx/1.21.0
date: Wed, 01 Apr 2026 16:28:08 GMT
content-type: text/html; charset=utf-8
content-length: 41959
Set-Cookie: nonce=mdp-1282809593; Max-Age=1343424; Path=/; Expires=Fri, 04 Sep 2026 04:12:11 GMT; HttpOnly
Set-Cookie: CSRF=d3d3dc98f08b20469080998cfc8427e...

Observed Item	Detail Found	Why It Matters to an Attacker
HTTP service	Port 80 open	Can be used to scan web vulnerabilities
HTTPS service	Port 443 open	Entry point for web-based attacks
Server software	Nginx/Apache	Helps attacker identify exploit targets
Version info	e.g. Apache 2.4.x	Enables known vulnerability lookup
Location	India / US	Helps in targeted attacks
Hosting provider	AWS / Cloudflare	Helps attacker understand infra

1. Which exposed service would be most valuable to an attacker, and why?

The most valuable exposed service is HTTP/HTTPS (ports 80/443) because web servers are publicly accessible and commonly targeted using vulnerabilities like SQL injection, XSS, or misconfigurations.

2. Why can an organisation not simply 'hide' its infrastructure from Shodan?

Organisations cannot hide because their systems must be publicly reachable to provide services. Shodan scans the internet continuously and indexes publicly exposed devices, making complete invisibility impossible.

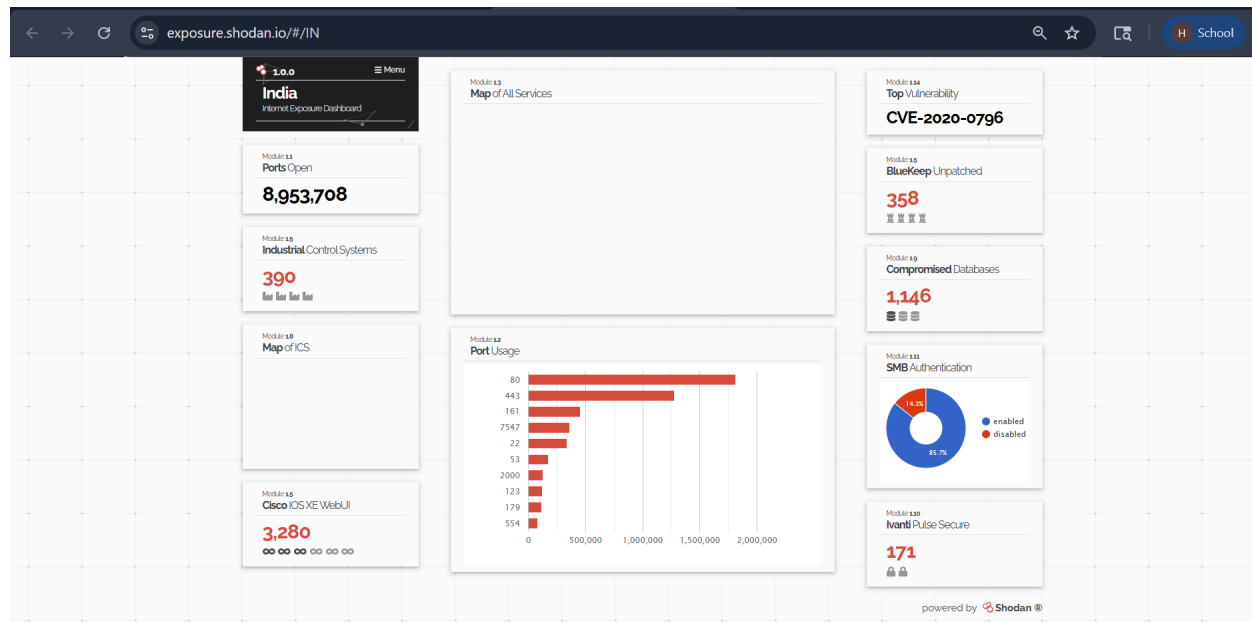
3. Does disclosing the version materially increase the attacker's advantage in this case? Explain

Yes, disclosing the software version significantly increases attacker advantage because it allows them to search for known vulnerabilities (CVEs) specific to that version and use ready-made exploits.

4. What single defensive change would most reduce the reconnaissance value of the exposed information you found?

The most effective defensive change would be **reducing banner information disclosure**, such as hiding server version details, as it limits reconnaissance and makes targeted attacks more difficult.

Task 2(b) Interpreting India's Internet Exposure using Shodan Observatory



Observation 1

A very high number of open ports (8,934,514) are exposed across systems in India.

Classification: Infrastructure exposure

Indicates a large attack surface, meaning many systems are reachable and potentially exploitable.

Observation 2

The most commonly used ports are 80 (HTTP) and 443 (HTTPS), followed by ports like 161, 22, and 7547.

Classification: Service exposure

Shows heavy reliance on web services and presence of management/remote access ports, which are frequent attack targets.

Observation 3

A significant number of Industrial Control Systems (ICS) (391) are exposed.

Classification: Infrastructure exposure

ICS systems are critical infrastructure (power, manufacturing), and exposure can lead to high-impact cyber attacks.

Observation 4

Vulnerabilities like CVE-2020-0796 and BlueKeep (357 unpatched systems) are still present.

Classification: Potential vulnerability signal

Indicates unpatched systems, which attackers can exploit using known vulnerabilities.

Observation 5

There are 1,151 compromised databases detected.

Classification: Potential vulnerability signal

Shows active data breaches or weak security configurations, making sensitive data accessible.

Technologies like Cisco IOS XE WebUI (3,287 instances) and Ivanti Pulse Secure (157 instances) are publicly visible.

Classification: Technology disclosure

TASK 3: Certificate Transparency Lookup and Subdomain Intelligence

crt.sh

Identity Search



Group by Issuer

CriteriaType: IdentityMatch: ILIKESearch: 'du.ac.in'

Sorry, your search results have been truncated.

It is not currently possible to sort and paginate large result sets efficiently, so only a random subset is shown below.

Please retry your search with [expired certificates excluded](#).

Certificates	crt.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities	Issuer Name
	9869908995	2023-07-10	2023-07-10	2023-10-08	www.alumniportal.sscbs.du.ac.in	alumniportal.sscbs.du.ac.in www.alumniportal.sscbs.du.ac.in	C=US, O=Let's Encrypt, CN=R3
	9868512922	2023-07-09	2023-07-09	2023-10-07	geography.du.ac.in	geography.du.ac.in www.geography.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9868512977	2023-07-09	2023-07-09	2023-10-07	geography.du.ac.in	geography.du.ac.in www.geography.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535651	2023-07-09	2023-07-09	2023-10-07	pbhcz0.geology.du.ac.in	pbhcz0.geology.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535638	2023-07-09	2023-07-09	2023-10-07	pbhcz0.geology.du.ac.in	pbhcz0.geology.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535424	2023-07-09	2023-07-09	2023-10-07	mac.du.ac.in	mac.du.ac.in www.mac.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535398	2023-07-09	2023-07-09	2023-10-07	mac.du.ac.in	mac.du.ac.in www.mac.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535241	2023-07-09	2023-07-09	2023-10-07	dusc.du.ac.in	dusc.du.ac.in www.dusc.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535353	2023-07-09	2023-07-09	2023-10-07	dusc.du.ac.in	dusc.du.ac.in www.dusc.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535202	2023-07-09	2023-07-09	2023-10-07	cs.du.ac.in	cs.du.ac.in www.cs.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866535009	2023-07-09	2023-07-09	2023-10-07	cs.du.ac.in	cs.du.ac.in www.cs.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"
	9866534742	2023-07-09	2023-07-09	2023-10-07	centenary.du.ac.in	centenary.du.ac.in	C=US, ST=TX, L=Houston, O="cPanel, Inc.", CN="cPanel, Inc. Certification Authority"

Subdomain Found	Certificate Issued By	Reconnaissance Value
geography.du.ac.in	cPanel Inc. Certification Authority	Reveals departmental infrastructure, may have weaker security controls
dusc.du.ac.in	cPanel Inc. Certification Authority	Indicates student-related system, potential target for data access
cs.du.ac.in	cPanel Inc. Certification Authority	Computer science department server, possible exposure of technical resources

1. Why do SSL certificates reveal subdomains even when those subdomains are not publicly advertised?

SSL/TLS certificates are logged in public Certificate Transparency logs to ensure transparency and prevent misuse. These logs include all associated subdomains, making them visible even if the organisation does not publicly advertise them.

2. What does the presence of subdomains like 'staging' or 'dev.' tell a potential attacker?

Subdomains like *staging* or *dev* indicate development or testing environments. These environments are often less secure, may contain debugging features, and are therefore attractive targets for attackers.

TASK 4: Risk Analysis and Exposure Classification

Information Found	Source (WHOIS / Shodan / crt.sh)	Exposure Classification	Risk Level	Justification
Domain registrar and registration details	WHOIS	Third-party dependency visibility	Low	This information is generally public and does not directly expose vulnerabilities, but it reveals dependency on external providers.
Name servers (DNS infrastructure)	WHOIS	Infrastructure disclosure	Medium	Reveals DNS infrastructure which attackers can target for DNS spoofing, hijacking, or DDoS attacks.
Limited/visible contact details	WHOIS	Contact exposure	Medium	Enables phishing and social engineering attacks by exposing individuals associated with the domain.
Open ports (80, 443, 22 etc.)	Shodan	Service exposure	High	Directly exposes entry points into the system, allowing attackers to attempt exploitation or brute-force attacks.
Hosting provider / ISP details	Shodan	Infrastructure disclosure	Medium	Helps attackers understand the hosting environment and tailor attacks specific to that provider.
Software banners (Apache/Nginx)	Shodan	Software version exposure	High	Discloses technologies and possibly versions, enabling attackers to use known exploits (CVEs).
Subdomains (geography.du.ac.in, cs.du.ac.in, dusc.du.ac.in)	crt.sh	Subdomain enumeration	High	Expands the attack surface by revealing additional systems that may have weaker security controls.
Certificate Authority (cPanel, Let's Encrypt)	crt.sh	Third-party dependency visibility	Low	Indicates external certificate providers but does not directly expose exploitable weaknesses.
Industrial/critical systems exposure (if observed)	Shodan	Infrastructure exposure	High	Exposure of critical systems can lead to severe consequences, including disruption of essential services.

1. What is the difference between passive and active reconnaissance? Give one concrete example of each from this lab.

Passive reconnaissance involves collecting information without directly interacting with the target system, while active reconnaissance involves direct interaction.

Example from your lab:

- Passive: Using WHOIS, Shodan, crt.sh
- Active: Port scanning using tools like Nmap (not performed here)

2. Why is the legal distinction between passive and active reconnaissance important for a security professional?

Passive reconnaissance is generally legal because it uses publicly available data, whereas active reconnaissance may be considered unauthorized access and can violate laws or organisational policies.

3. A penetration tester discovers that a company's WHOIS record lists a named employee's personal email. Why is this a security risk beyond just privacy?

It enables targeted phishing and social engineering attacks. Attackers can impersonate trusted entities and exploit the individual to gain access to organisational systems, making it a security risk beyond privacy concerns.

4. Based on all three tasks, which source (WHOIS, Shodan, or crt.sh) produced the most useful reconnaissance information about your domain, and why?

Shodan provided the most useful reconnaissance information because it revealed live system data such as open ports, running services, and software details, which can be directly used for identifying vulnerabilities and planning attacks.

5. Which of your findings would be most useful to:
a. a phishing-oriented attacker,
b. an infrastructure-focused attacker,
c. An attacker attempting technology fingerprinting?
Justify your answer.

a. Phishing-oriented attacker

Contact information and email-related subdomains (e.g., mail/webmail) are most useful because they help craft targeted phishing attacks.

b. Infrastructure-focused attacker

Open ports, hosting provider, and IP-related data from Shodan are most useful as they reveal the system architecture and entry points.

c. Technology fingerprinting attacker

Software banners and version information are most useful because they allow attackers to identify specific technologies and exploit known vulnerabilities.